Hackpad Submission Checklist

From the official page blueprint.hackclub.com/hackpad/submitting, read 18 Oct 2026. Certain

Hard rules

Rule	TechPad
Through-hole Seeed XIAO RP2040 as the main MCU	✓
PCB ≤ 100 × 100 mm, 2 layers	✓ 92 × 87, 2 layers
Fewer than 16 inputs	✓ 12 keys + 1 encoder
Approved parts only	✓ kit parts only
Case fully 3D-printed (no acrylic or laser-cut parts)	✓ planned
Case fits 200 × 200 × 100 mm	✓ 98 × 93 × about 30 with keycaps
Original firmware (QMK, KMK or ZMK based)	○ to do

What's done and what's left

Item	Status
PCB files (pcb.kicad_pro, .kicad_sch, .kicad_pcb, .kicad_dru)	✓ Done, 0 ERC and 0 DRC errors
Gerbers + drill files	✓ Exported; zip them as gerbers.zip
Case STEP files (each part + full assembly with the PCB)	○ You, in FreeCAD
Firmware source + firmware.uf2	○ Claude
Screenshots (schematic, PCB, case, whole pad)	○ You
README.md with the BOM	○ Claude can draft it once the screenshots exist

Folder layout to hand in

```
TechPad/
├── CAD/          TechPad.step: whole case + PCB (a blank box is fine), TechPad_Case.FCStd
├── PCB/          pcb.kicad_pro, pcb.kicad_sch, pcb.kicad_pcb (+ pcb.kicad_dru)
├── Firmware/     QMK keyboard folder (keyboard.json, keymap, custom matrix code)
├── Production/   gerbers.zip, Frame.step, Plate.step, Tray.step, Knob.step, firmware.uf2
└── README.md
```

README must include

- A picture of the whole hackpad (render or photo)
- A schematic screenshot
- A PCB screenshot
- A case screenshot showing how it fits together
- The BOM (03_BOM_and_Hardware)

Be honest in the README. Likely Say that the firmware and case fits are untested because the parts aren't here yet. Reviewers expect this, and it's better than a claim that turns out to be wrong.